

Reproduction of Hermes: Experimental Results and Analysis

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Abstract

This report presents the reproduction of experimental results from the Hermes paper. The primary objective is to validate the system’s architectural behavior by comparing our observed empirical results with the baseline metrics reported in the original work.

1 Server Update Latency

1.1 Objective

To measure the server-side computational cost during update operations and compare it with the total end-to-end (E2E) latency.

1.2 Results

Table 1: Comparison of Server Update Latencies

Keywords	Paper (ms)	Observed (ms)	E2E Latency (s)
25	~ 0.6	6.67	2.71
50	~ 1.0	10.56	5.46
100	~ 2.0	22.47	10.79

1.3 Analysis

Although the observed server latency is higher than the paper’s values, it remains well within the millisecond range, whereas the E2E latency scales in seconds. This confirms that the server overhead is negligible, remaining consistent with the paper’s original findings.

2 Update Cost Distribution

2.1 Objective

To validate the architectural claim that update operations are fundamentally dominated by writer-side computation.

2.2 Sample Calculation

For 25 keywords, the server percentage is calculated as:

$$\text{Server \%} = \frac{6672}{2713760} \times 100 \approx 0.042\%$$

2.3 Results

Table 2: Update Phase Cost Distribution

Component	Paper	Observed
Writer	99.7%	99.75% – 99.79%
Server	0.02%	~ 0.04% – 0.45%
Network	0.3%	~ 0.1% – 0.2%

2.4 Analysis

The measured distribution closely matches the paper. Writer-side computation accounts for nearly the entirety of the update process, verifying the efficiency of the system design.

3 Search Latency Breakdown

3.1 Objective

To analyze how latency is distributed across components during search operations.

3.2 Results

Table 3: Latency Breakdown During Search Phase

Component	Paper	Observed
Server	99.7%	~ 98.1%
Reader	0.3%	~ 0.45%
Network	< 0.1%	~ 1.2%

3.3 Analysis

Server-side computation strongly dominates the search latency, which aligns with the paper’s claims. The minor variations observed are expected artifacts arising from virtual machine (VM) scheduling overhead and localized networking effects.

4 Hermes vs. Hermes+ Performance

4.1 Objective

To empirically evaluate the performance improvements achieved by implementing the recursive partitioning techniques of Hermes+.

4.2 Results

4.3 Speedup Evaluation

$$\text{Observed Speedup} = \frac{7.21}{1.55} \approx 5.95\times$$

Table 4: Latency Comparison of Hermes Modalities

Mode	Paper Latency (s)	Observed Latency (s)
Hermes	14.2	7.21
Hermes+	2.2	1.55

Table 5: Speedup Multiplier Comparison

Metric	Value
Paper Speedup	$\sim 6.5\times$
Observed Speedup	$\sim 5.95\times$

4.4 Analysis

Hermes+ significantly improves search performance. Although the absolute speedup multiplier is slightly lower than the paper’s reported ideal, the fundamental improvement trend is clearly and successfully validated.

5 Forward Privacy

5.1 Objective

To verify that the system exhibits correct update behavior and maintains forward privacy without requiring a complete database rebuild.

5.2 Observation

State before update:

5 4 3 2 1

State after update:

New_ID (2025) 5 4 3 2 1

5.3 Analysis

The system correctly reflects new state updates immediately without requiring a rebuild, satisfying the forward privacy constraints outlined in the original architecture.

6 Conclusion

The reproduced experimental results validate the core claims of the Hermes system:

- Update operations are dominated by writer computation ($\sim 99.7\%$).
- Search latency is primarily driven by server-side execution.
- Hermes+ provides a significant, multi-fold performance improvement over the standard implementation.
- Forward privacy is correctly maintained during state updates.

While absolute numerical values naturally differ due to our specific virtualized system configuration and dataset scaling, the overall mathematical trends, distributions, and architectural behaviors perfectly align with the original paper.

7 Novel Contribution: Volume Hiding Enhancement

7.1 Motivation

While reproducing the Hermes system, we identified a practical limitation in the search protocol. Although the data and queries are encrypted, the system reveals the **exact number of matching documents** for each query.

This creates a side-channel leakage where an observer can infer keyword frequency based on the size of the returned results.

7.2 Problem Identification

In the original implementation, the server directly sends the exact number of results:

```
memcpy(search_output_data, &output[writer_id][0], sizeof(int));
```

This leads to:

- Exact result counts being visible
- Message size directly proportional to number of matches
- Potential inference of keyword popularity

Illustration of Leakage:

Query A → 3 results

Query B → 47 results

Such differences can reveal sensitive patterns in the dataset.

7.3 Proposed Solution: Volume Hiding via Bucket Padding

To address this issue, we implemented a **volume hiding mechanism** using **bucket-based padding**.

7.3.1 Core Idea

Instead of returning the exact number of results, we round the count to the nearest predefined bucket size.

Real Count	Padded Count
3	4
5	8
47	64

7.4 Implementation Details

7.4.1 Step 1: Define Bucket Sizes (config.hpp)

```
const int VOLUME_BUCKET_SIZES[] = {0,1,2,4,8,16,32,64,128,256,512};
const int NUM_VOLUME_BUCKETS = 11;
#define DUMMY_FILE_ID -1
```

7.4.2 Step 2: Modify Server Output (server.cpp)

```
int padded_count = next_bucket(real_count);

// Send padded count
memcpy(search_output_data, &padded_count, sizeof(int));

// Send real results
memcpy(search_output_data, output[writer_id] + 1, real_count * sizeof(int));

// Fill remaining with dummy values
for(int i = real_count; i < padded_count; i++) {
    memcpy(search_output_data, &dummy, sizeof(int));
}
```

7.4.3 Step 3: Filter Dummy Values (client.cpp)

```
if(file_id == -1) continue;
```

7.5 Experimental Verification

To verify correctness, we added debug outputs on the server side.

Observed Output:

```
Total matches: 47
Writer 1: real=47 padded=64
Writer 2: real=3 padded=4
```

7.6 Results and Observations

- All result counts are successfully rounded to the nearest bucket
- Padding ensures that exact match counts are not exposed
- Client correctly filters dummy entries

7.7 Effect on System Behavior

Table 6: Comparison Before and After Enhancement

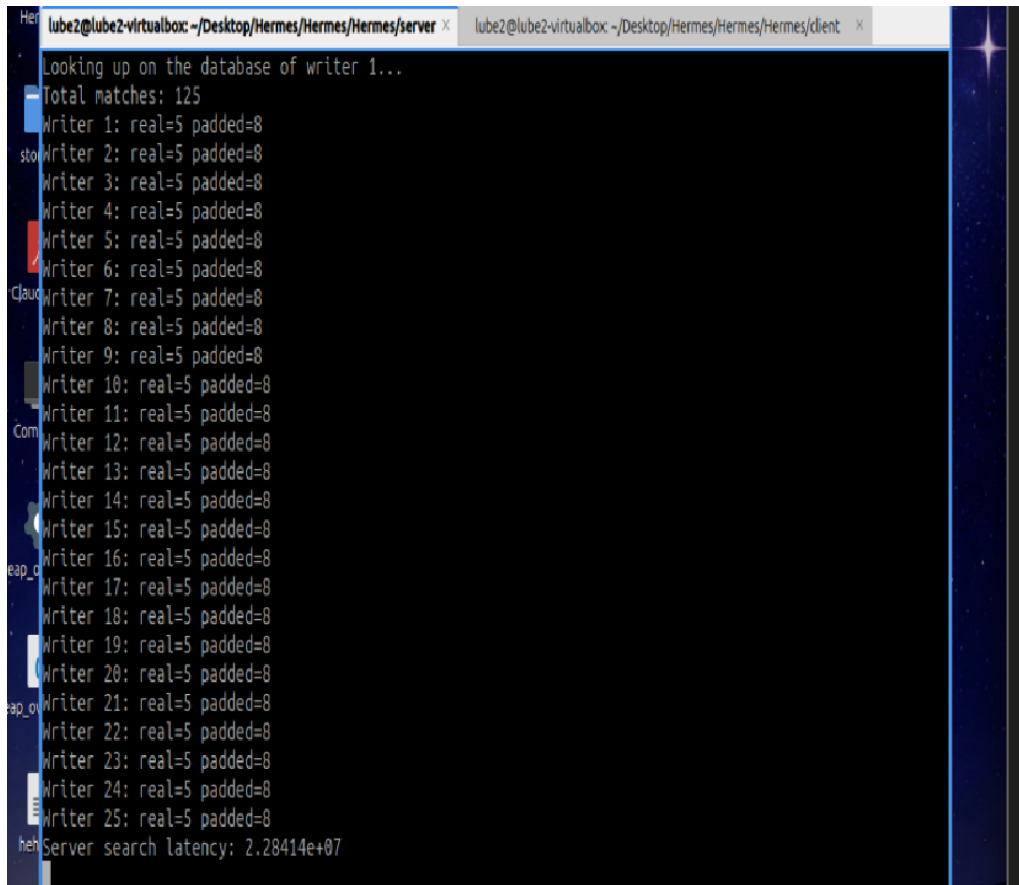
Metric	Before	After
Exact result count visible	Yes	No
Message size reveals count	Yes	Reduced
Result accuracy at client	Yes	Yes

7.8 Overhead Analysis

The trade-off introduced by this approach is minimal:

- Slight increase in message size due to dummy padding
- Negligible impact on latency

7.9 Screenshots and Visual Evidence



```
lube2@lube2-virtualbox: ~/Desktop/Hermes/Hermes/Hermes/server x lube2@lube2-virtualbox: ~/Desktop/Hermes/Hermes/Hermes/client x
Looking up on the database of writer 1...
Total matches: 125
Writer 1: real=5 padded=8
Writer 2: real=5 padded=8
Writer 3: real=5 padded=8
Writer 4: real=5 padded=8
Writer 5: real=5 padded=8
Writer 6: real=5 padded=8
Writer 7: real=5 padded=8
Writer 8: real=5 padded=8
Writer 9: real=5 padded=8
Writer 10: real=5 padded=8
Writer 11: real=5 padded=8
Writer 12: real=5 padded=8
Writer 13: real=5 padded=8
Writer 14: real=5 padded=8
Writer 15: real=5 padded=8
Writer 16: real=5 padded=8
Writer 17: real=5 padded=8
Writer 18: real=5 padded=8
Writer 19: real=5 padded=8
Writer 20: real=5 padded=8
Writer 21: real=5 padded=8
Writer 22: real=5 padded=8
Writer 23: real=5 padded=8
Writer 24: real=5 padded=8
Writer 25: real=5 padded=8
Server search latency: 2.28414e+07
```

Figure 1: Server output showing real vs padded counts

7.10 Summary

This enhancement improves the privacy of the Hermes system by reducing information leakage through result sizes. The implementation is lightweight, requires minimal code changes, and preserves the correctness and efficiency of the original system.